

The MOND Interpolation Function as a Bose–Einstein Partition Function over Gravitational Entropy Modes: A Complexified Extension of the Einstein–Hilbert Action

Anderson M. Rodriguez*

Independent Researcher, Athens, GA, USA

amr28693@uga.edu

April 22, 2026

Abstract

We extend the Einstein–Hilbert action to a complexified form $\mathcal{L}_C = \mathcal{L}_R + i\mathcal{L}_I$ in which the imaginary sector encodes entropy flow through a scalar potential $\theta(x)$ with kinetic Lagrangian $\mathcal{L}_I = \alpha g^{\mu\nu} \nabla_\mu \theta \nabla_\nu \theta$. The complexified field equations carry a coupling of $16\pi G$; a real-sector projection with equal-weight measure over the imaginary axis recovers the standard Einstein equations with $8\pi G$ and leaves a decoupled entropy equation $\square \theta = 0$. In the static spherically symmetric case this yields $\theta(r) = \sqrt{GM/a_0} r^{-1}$, identical in form to the MOND parameter $\sqrt{g_{\text{bar}}/a_0}$. We then show that the empirical MOND interpolation function $g_{\text{obs}} = g_{\text{bar}}/(1 - e^{-\theta})$ is algebraically a Bose–Einstein partition function over a single bosonic mode of dimensionless energy θ , so that the mass discrepancy decomposes as $D \equiv g_{\text{obs}}/g_{\text{bar}} = 1 + n_{\text{BE}}(\theta)$. The baryon-to-entropy crossover threshold at $n_{\text{BE}} = 1$ occurs at $\theta = \ln 2$ (one bit of information), predicting a transition at $g_{\text{bar}}/a_0 = (\ln 2)^2 = 0.4805$ with no free parameters aside from a_0 . Testing against 2,693 data points in 153 SPARC galaxies reproduces the mass discrepancy with a residual standard deviation of 0.133 dex, consistent with the known observational error budget of the radial acceleration relation and leaving no room for intrinsic scatter beyond measurement precision. The present work is situated within the thermodynamic interpretation of gravitational field equations established by Jacobson and Padmanabhan, and produces a concrete observable consequence in the low-acceleration regime of galactic dynamics.

1 Introduction

The radial acceleration relation [1, 2] establishes a tight, galaxy-independent correlation between the observed centripetal acceleration g_{obs} and the Newtonian acceleration g_{bar} computed from the baryonic mass distribution. Across more than four decades in g_{bar} and 153

*amr28693@uga.edu

disk galaxies in the SPARC sample, the data are well fit by the interpolation function [3, 4]

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{1 - \exp\left(-\sqrt{g_{\text{bar}}/a_0}\right)}, \quad (1)$$

with $a_0 \simeq 1.2 \times 10^{-10} \text{ m s}^{-2}$ [5, 6]. Equation (1) reduces to Newtonian behavior for $g_{\text{bar}} \gg a_0$ and to the Milgrom deep-MOND limit [7] $g_{\text{obs}} \rightarrow \sqrt{g_{\text{bar}} a_0}$ for $g_{\text{bar}} \ll a_0$, and has been used as an empirical template rather than derived from an action principle. We note at the outset that recent work has ruled out MOND as a modified gravitational force law on cosmological scales at 3.3σ [8]. The present paper does not propose a modified force law. The Einstein equations are recovered exactly (Section 2.3), and the gravitational force remains $1/r^2$ at all scales; the mass discrepancy in galactic dynamics arises from the thermal occupation of entropy modes, not from a modification of dynamics. Nevertheless, the exponential $e^{-\sqrt{g_{\text{bar}}/a_0}}$ has no known origin in any gravitational Lagrangian.

Parallel developments in the thermodynamic approach to gravitation suggest that such a derivation ought to be available. Jacobson [9] recovered the Einstein equations as an equation of state from the Clausius relation $\delta Q = T \delta S$ imposed on local Rindler horizons. Padmanabhan [10] extended the thermodynamic interpretation to a wide class of gravitational actions, and Verlinde [11] proposed gravity itself as an entropic force emergent from holographic entropy. Banerjee and Majhi [12] showed directly that the gravitational action is proportional to statistical-mechanical entropy, with the black hole relation $S = E/2T$ emerging from the Komar energy and Hawking temperature.

The present paper derives the functional form of the galactic mass discrepancy [13, 14] from a complexified extension of the Einstein–Hilbert action [15]. It is organized around a single claim with two components.

The first is algebraic and immediate: Equation (1) is not merely thermal in flavor; it is a Bose–Einstein partition function over a single mode whose dimensionless energy is $\theta = \sqrt{g_{\text{bar}}/a_0}$, so that the mass discrepancy decomposes exactly as $D \equiv g_{\text{obs}}/g_{\text{bar}} = 1 + n_{\text{BE}}(\theta)$ with $n_{\text{BE}}(\theta) = 1/(e^\theta - 1)$.

The second is dynamical: this θ arises as the static spherically symmetric solution of a massless scalar equation $\square\theta = 0$ governing an entropy sector that decouples from the metric when the Einstein–Hilbert action is extended to a complexified form and projected onto its real observational slice. The threshold at $n_{\text{BE}} = 1$ occurs at $\theta = \ln 2$ —one bit of Shannon entropy—and predicts a transition at $g_{\text{bar}}/a_0 = (\ln 2)^2 = 0.4805$ with no free parameters beyond a_0 itself.

We adopt the standpoint that the radial acceleration relation is a specific consequence of the complexified Einstein–Hilbert action constructed here. The framework, a complexified Einstein–Hilbert action with a postulated equal-weight projection onto the real sector, produces observable consequences wherever the entropy field is sourced; the radial acceleration relation is the first worked consequence because the static spherically symmetric solution of $\square\theta = 0$ is elementary and the data are clean. We find that the structure generalizes. The framework reframes the empirical MOND phenomenology as the thermal occupation of gravitational entropy modes, makes the dark-matter contribution a bookkeeping quantity rather than a material one, and locates the baryon-to-entropy transition at an information-theoretically natural scale.

The paper proceeds as follows. Section 2 constructs the complexified Einstein–Hilbert action and its variational structure, introduces the real-sector projection, and recovers the $8\pi G$ coupling from an inner-product measure on the complexified manifold. Section 3 derives the decoupled entropy equation $\square\theta = 0$ and its static spherically symmetric solution. Section 4 identifies the resulting entropy potential with the MOND parameter. Section 5 shows algebraically that the MOND interpolation function is a Bose–Einstein partition function. Section 6 locates the one-bit threshold. Section 7 tests the prediction against the full SPARC RAR sample. Section 8 places the result in the context of thermodynamic gravity and identifies the open questions, and Section 9 concludes. Appendix A contains numerical visualizations of the projection mechanism.

2 The Complexified Einstein–Hilbert Action

2.1 Complex Lagrangian Density

We postulate a Lagrangian density over a complexified manifold $\mathcal{M}_{\mathbb{C}} \cong \mathcal{M} \times \mathcal{E}$, where $\mathcal{M}$ is ordinary Lorentzian spacetime with coordinates x^μ and $\mathcal{E}$ is a one-dimensional entropy axis parameterized by a coordinate τ . The auxiliary axis $\mathcal{E}$ is not a spacetime dimension and carries no metric or independent dynamics, but instead it is a statistical axis labeling entropy-degenerate sectors; only its measure enters the physics through the projection defined in Section 2.3.

On $\mathcal{M}_{\mathbb{C}}$ we take

$$\mathcal{L}_{\mathbb{C}} = \mathcal{L}_R + i \mathcal{L}_I, \quad (2)$$

where $\mathcal{L}_R = R$ is the Ricci scalar on $\mathcal{M}$ in standard General Relativity conventions [16] and

$$\mathcal{L}_I = \alpha g^{\mu\nu} \nabla_\mu \theta \nabla_\nu \theta \quad (3)$$

is the kinetic Lagrangian of a scalar entropy potential $\theta(x)$ with real normalization α . The imaginary unit i in Eq. (2) plays the role of a sector label: real and imaginary parts correspond to observable curvature and entropy-sector dynamics respectively. We use capital subscripts R and I to denote them, with the subscripts being sector labels, not tensor indices.

This construction differs from the algebraic complexification of a vector space [17] in that the imaginary sector carries independent physical degrees of freedom (i.e., the scalar entropy potential $\theta(x)$) whose dynamics are fixed by variation of $S_{\mathbb{C}}$ and whose observable consequences emerge only after projection onto the real sector.

The quadratic form of $\mathcal{L}_I$ is the simplest Lorentz-invariant kinetic term for a scalar [18]; in the analogy with non-equilibrium statistical mechanics [12], $\nabla_\mu \theta$ plays the role of a local entropy current. The full complex action is

$$S_{\mathbb{C}} = \int d^4x \sqrt{-g} \mathcal{L}_{\mathbb{C}} = \int d^4x \sqrt{-g} (\mathcal{L}_R + i \mathcal{L}_I). \quad (4)$$

The entropy associated with the scalar field $\theta(x)$ is understood in the same coarse-grained and emergent sense employed in thermodynamic derivations of the gravitational field equations, not as a microscopic state count of an underlying particle theory. In particular, θ parameterizes an information-theoretic entropy defined operationally through thermodynamic

response, of the same character as Jacobson’s horizon entropy derived from the Clausius relation [9] and the surface degrees of freedom appearing in Padmanabhan’s thermodynamic formulation of gravity [10], though defined here through the projection mechanism rather than on a horizon. No assumption is made here about a specific microscopic completion; the entropy is defined by its role in the projection from the complexified to the observable sector.

2.2 Variation and the $16\pi G$ Complexified Field Equations

Variation of (4) with respect to $g^{\mu\nu}$ is linear in the Lagrangian, so real and imaginary sectors vary independently,

$$\delta S_{\mathbb{C}} = \delta S_R + i \delta S_I = 0. \quad (5)$$

The real sector yields the standard Einstein–Hilbert variation and the imaginary sector yields

$$\delta S_I = \alpha \int d^4x \sqrt{-g} \delta g^{\mu\nu} \nabla_\mu \theta \nabla_\nu \theta, \quad (6)$$

so that the complexified field equations take the form

$$G_{\mathbb{C}}^{\mu\nu} = 16\pi G T_{\mathbb{C}}^{\mu\nu}, \quad G_{\mathbb{C}}^{\mu\nu} = G_R^{\mu\nu} + i G_I^{\mu\nu}, \quad T_{\mathbb{C}}^{\mu\nu} = T_R^{\mu\nu} + i T_I^{\mu\nu}. \quad (7)$$

The coupling $16\pi G$ —twice the standard Einstein coupling—is a direct consequence of integrating both real and imaginary kinetic contributions on equal footing. The factor of two is what the projection onto the real sector must remove.

2.3 Real-Sector Projection from an Equal-Weight Measure

Define a real-sector projection

$$\mathbb{P}_{\text{Re}}[A_{\mathbb{C}}^{\mu\nu}] := \int_{\mathcal{E}} d\tau w(\tau) A_{\mathbb{C}}^{\mu\nu}(x, \tau), \quad (8)$$

where $w(\tau)$ is a normalized window function on the entropy axis, centred at $\tau = 0$ and symmetric under reflection $\tau \mapsto -\tau$. For tensor fields satisfying a Schwarz reflection [19] symmetry on $\mathcal{E}$, namely $A_{\mathbb{C}}^{\mu\nu}(x, \tau) = A_{\mathbb{C}}^{\mu\nu}(x, -\tau)^*$, this projection selects the real part of the field evaluated on $\tau = 0$:

$$\mathbb{P}_{\text{Re}}[A_{\mathbb{C}}^{\mu\nu}] = \text{Re}[A_{\mathbb{C}}^{\mu\nu}(x, 0)]. \quad (9)$$

We state the projection’s normalization as a single postulate whose physical content is equipartition between the real and imaginary sectors.

Postulate 1 (Equal-Weight Measure). *The measure $w(\tau)$ on the entropy axis $\mathcal{E}$ assigns equal statistical weight to the real and imaginary sectors of $\mathcal{M}_{\mathbb{C}}$; equivalently, the expectation of a tensor observable with independent real and imaginary contributions reduces to $\mathbb{P}_{\text{Re}}[A_R^{\mu\nu} + i A_I^{\mu\nu}] = \frac{1}{2} A_R^{\mu\nu}$ on the reflection-symmetric class.*

The equal-weight measure is the unique normalized measure on $\mathcal{E}$ invariant under the conjugation symmetry $\tau \mapsto -\tau$ of the complexified manifold; Postulate 1 is therefore the statement that the projection respects the symmetry of the space on which it acts.

The postulate reformulates the equal-weighting condition as a structural property of the measure rather than an ad-hoc factor. It plays a role analogous to the Clausius assumption in Jacobson’s derivation and the holographic-entropy assumption in Verlinde’s: each construction requires one thermodynamic input beyond which the gravitational dynamics follow. We return in Section 8 to the question of whether the equal-weight measure can be obtained from a deeper condition.

Applying $\mathbb{P}_{\text{Re}}$ to Eq. (7) gives

$$G_R^{\mu\nu} = 16\pi G \mathbb{P}_{\text{Re}}[T_C^{\mu\nu}] = 16\pi G \cdot \frac{1}{2} T_R^{\mu\nu}, \quad (10)$$

so that

$$\boxed{G_R^{\mu\nu} = 8\pi G T_R^{\mu\nu}}. \quad (11)$$

The Einstein equations with their standard coupling are recovered on the real observational slice. The factor $\frac{1}{2}$ is the statistical trace over the entropy-degenerate sector, analogous to a partial trace over environmental degrees of freedom in quantum statistical mechanics [20].

The projection is formally analogous to two established constructions. Wick rotation $t \mapsto -i\tau$ in quantum field theory [18] analytically continues real-time dynamics to a Euclidean slice; our projection selects the real slice of an already complexified manifold without analytic continuation. Partial traces in statistical mechanics [20] integrate out environmental degrees of freedom to yield a reduced density matrix; $\mathbb{P}_{\text{Re}}$ integrates out the entropy axis τ to yield the reduced, observable curvature. In both analogies the operation is natural; the postulated equal weighting is the physical content beyond the formal construction.

3 The Entropy Field Equation and Its Static Solution

While the projection (11) removes the imaginary sector from the observable metric dynamics, the imaginary variation itself yields a decoupled field equation for the entropy potential. Varying (3) with respect to θ and using Stokes’ theorem [16] gives

$$\boxed{\square \theta = 0}, \quad (12)$$

where $\square = g^{\mu\nu} \nabla_\mu \nabla_\nu = (-g)^{-1/2} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu)$ is the d’Alembertian on $\mathcal{M}$. The entropy potential is a massless scalar propagating on the real spacetime produced by Eq. (11), and it does not back-react on the metric to the order considered here.¹

3.1 Static Spherically Symmetric Solution

In the static, asymptotically flat, spherically symmetric case the d’Alembertian reduces to the flat-space Laplacian at leading order in the Newtonian limit [16], and Eq. (12) becomes

¹Back-reaction enters if the kinetic term (3) is permitted to contribute a real stress-energy after projection. Postulate 1 suppresses this at leading order; see Section 8 for the open status of the non-spherical and time-dependent cases.

the Laplace equation

$$\nabla^2 \theta(r) = 0, \quad r > 0. \quad (13)$$

Imposing the regularity and fall-off conditions $\theta(r) \rightarrow 0$ as $r \rightarrow \infty$ and requiring sourcing by a central mass selects the $1/r$ branch [21]:

$$\theta(r) = \frac{B}{r}, \quad (14)$$

with B a constant of dimension length determined by the source mass and the acceleration scale intrinsic to the entropy sector.

Equation (14) is the simplest nontrivial solution of $\square \theta = 0$ in the presence of a compact source; it extrapolates smoothly from the Newtonian far-field regime to the regime where entropy-mode occupation becomes non-negligible, which will be shown in Sec. 4 to be the regime probed by galaxy rotation curves.

4 Identification with the MOND Parameter

The baryonic gravitational acceleration at radius r from a spherical mass M is $g_{\text{bar}} = GM/r^2$ [5]. The dimensionless empirical MOND parameter appearing in the RAR fit, defined in the Introduction by Eq. (1) is

$$\theta_{\text{MOND}}(r) = \sqrt{\frac{g_{\text{bar}}(r)}{a_0}} = \sqrt{\frac{GM}{a_0}} \frac{1}{r}. \quad (15)$$

Equating this with the static solution (14) fixes the integration constant:

$$\boxed{B = \sqrt{\frac{GM}{a_0}}}. \quad (16)$$

The entropy potential derived from the complexified action is the MOND parameter. The identification is exact, involves no fitting, and follows from Eq. (12) with standard boundary conditions.

The constant B has dimensions of length and defines the *MOND radius*

$$r_M \equiv B = \sqrt{\frac{GM}{a_0}}, \quad (17)$$

the scale at which entropy-mode occupation becomes of order unity.

For a Milky Way-mass galaxy ($M \sim 5 \times 10^{10} M_\odot$) one finds $r_M \simeq 7.6$ kpc, consistent with the observed onset of flat rotation curves [6]. The acceleration scale a_0 enters the framework through this identification; within the scope of the present paper its numerical value is an empirically determined constant, a status it shares with MOND itself. The present framework does, however, assign a_0 a physical meaning it previously lacked: it is the characteristic entropy-temperature scale of the gravitational entropy sector, fixing the normalization of the entropy potential $\theta(r)$ through the boundary condition (16).

5 The Bose–Einstein Partition-Function Structure

5.1 Algebraic Identity

With θ identified as the static spherical solution of the entropy field equation, the empirical MOND interpolation function (1) can be rewritten without any approximation:

$$\frac{g_{\text{obs}}}{g_{\text{bar}}} = \frac{1}{1 - e^{-\theta}} = \frac{e^{\theta}}{e^{\theta} - 1} \quad (18)$$

$$= 1 + \frac{1}{e^{\theta} - 1} \quad (19)$$

$$= 1 + n_{\text{BE}}(\theta), \quad (20)$$

where

$$\boxed{n_{\text{BE}}(\theta) = \frac{1}{e^{\theta} - 1}} \quad (21)$$

is the Bose–Einstein mean occupation number for a single mode of dimensionless energy θ at unit temperature [22].

5.2 Physical Content

The mass discrepancy defined by $D \equiv g_{\text{obs}}/g_{\text{bar}}$ decomposes as

$$D = \underbrace{1}_{\text{baryonic}} + \underbrace{n_{\text{BE}}(\theta)}_{\text{entropy modes}}. \quad (22)$$

The “dark matter” contribution at each radius is the mean occupation number of gravitational entropy modes thermally populated according to Bose–Einstein statistics, with the entropy potential $\theta(r)$ serving as the dimensionless energy parameter. Since $\theta = \sqrt{g_{\text{bar}}/a_0}$, large θ corresponds to the high-acceleration Newtonian regime, small θ to the low-acceleration deep-MOND regime, and the threshold at $\theta = \ln 2$ marks the scale at which entropy modes first contribute equally to the baryonic term. This yields three regimes.

For $\theta \gg 1$ (strong gravity, small radius) the occupation $n_{\text{BE}} \rightarrow e^{-\theta}$ is exponentially suppressed; entropy modes are frozen out and Newtonian gravity is sufficient. For $\theta \ll 1$ (weak gravity, large radius) the Rayleigh–Jeans tail [22] $n_{\text{BE}} \rightarrow 1/\theta$ dominates, producing a large mass discrepancy consistent with the deep-MOND regime. At the intermediate scale $\theta = \ln 2$ the occupation equals exactly unity: $n_{\text{BE}}(\ln 2) = 1/(2 - 1) = 1$ and the mass discrepancy is $D = 2$. The value $D = 2$ at $\theta = \ln 2$ follows directly from $n_{\text{BE}}(\ln 2) = 1/(e^{\ln 2} - 1) = 1/(2 - 1) = 1$.

The interpretation is thermodynamic. The partition function $Z(\theta) = 1/(1 - e^{-\theta})$ in Eq. (18) is the partition function of a single bosonic mode [22], and the ratio $g_{\text{obs}}/g_{\text{bar}}$ is identical to $Z(\theta)$. The mass discrepancy is therefore the full partition function of the entropy sector at the local value of θ , with the constant-1 contribution corresponding to the ground state and $n_{\text{BE}}(\theta)$ to the excited-state occupation. This makes the exponential in the RAR

fit not phenomenological but the Boltzmann factor of a single-mode thermodynamic system defined on the entropy axis.

The temperature of the system has a natural identification with the Unruh temperature associated with the acceleration scale a_0 . The Unruh relation [23] $k_B T = \hbar a / (2\pi c)$ at $a = a_0$ defines a characteristic temperature T_0 ; expressing θ in Unruh units yields $\theta = \varepsilon / (k_B T_0)$, so that the Bose–Einstein form $1/(e^\theta - 1)$ is the thermal occupation of modes of energy $\varepsilon = k_B T_0 \sqrt{g_{\text{bar}}/a_0}$. The thermodynamic framing thus connects smoothly to the Rindler-horizon thermodynamics of Jacobson and to Padmanabhan’s horizon-based derivations, with θ playing the role of a local inverse temperature.

6 The Threshold at One Bit and its Observational Crossover

The condition $n_{\text{BE}} = 1$ —exactly one quantum of entropy-mode excitation—gives

$$\frac{1}{e^\theta - 1} = 1 \quad \implies \quad e^\theta = 2 \quad \implies \quad \theta = \ln 2. \quad (23)$$

Since $\theta = \sqrt{g_{\text{bar}}/a_0}$, the threshold in terms of the baryonic acceleration is

$$\boxed{\left. \frac{g_{\text{bar}}}{a_0} \right|_{n_{\text{BE}}=1} = (\ln 2)^2 = 0.4805.} \quad (24)$$

This is a zero-parameter prediction: the threshold depends only on the transcendental constant $\ln 2$ and the MOND acceleration scale a_0 , both of which are already fixed.

Viewed together, the two lenses make the significance of $\theta = \ln 2$ unavoidable. *Information-theoretically*, $\ln 2$ nats is the Shannon entropy of a single bit: the threshold is the locus at which the entropy sector accommodates one bit of excitation. *Structurally*, the $\frac{1}{2}$ in the real-sector projection (Postulate 1) and the $\ln 2$ in the threshold are dual expressions of the same content: the projection selects one half of the complexified geometry, and by Landauer’s Principle the entropic cost of that binary selection is $k_B \ln 2$ per mode [24]. Tracing out half the complexified degrees of freedom at the $\tau = 0$ observational slice costs one bit per entropy mode; the threshold is where that bit first appears in the observed gravitational dynamics.

At $\theta = \ln 2$, $D = 2$ and $g_{\text{obs}} = 2g_{\text{bar}}$. We refer to this scale as the one-bit threshold rather than the crossover to distinguish the specific value $\theta = \ln 2$, which has information-theoretic content, from the broader transition zone between Newtonian and deep-MOND regimes that the term crossover typically denotes in the observational literature. This is the acceleration at which the baryonic and entropy contributions are equal, and it marks the transition from the ordinary gravitational regime ($g_{\text{bar}} \gtrsim a_0$, entropy modes frozen out) to the deep-MOND regime ($g_{\text{bar}} \ll a_0$, Rayleigh–Jeans tail dominant).

7 Empirical Validation

7.1 Dataset and Procedure

We test Eq. (22) against the radial acceleration relation dataset of Lelli et al. [2], consisting of $N = 2,693$ individual measurements of g_{bar} and g_{obs} across 153 SPARC galaxies. No

subsample selection is applied beyond the published SPARC RAR cuts: all 2,693 published points are used. For each point i we compute

$$\theta_i = \sqrt{g_{\text{bar},i}/a_0}, \quad D_{\text{pred},i} = 1 + \frac{1}{e^{\theta_i} - 1}, \quad D_{\text{obs},i} = g_{\text{obs},i}/g_{\text{bar},i}, \quad (25)$$

using $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$. The calculation involves no free parameters beyond a_0 , which is held fixed at its RAR best-fit value throughout.

7.2 Residuals

The logarithmic residuals $\Delta_i = \log_{10} D_{\text{obs},i} - \log_{10} D_{\text{pred},i}$ have

$$\begin{aligned} \text{Mean residual:} & \quad -0.004 \text{ dex} \\ \text{Std. deviation:} & \quad 0.133 \text{ dex} \\ \text{RMS residual:} & \quad 0.133 \text{ dex} \end{aligned}$$

The scatter is to be compared in magnitude with the intrinsic scatter of the RAR discovered qualitatively by McGaugh et al. [1], and quantitatively reported by Lelli et al. [2], which is ~ 0.13 dex. The Bose–Einstein model reproduces the full systematic structure of the data; the residuals are consistent with intrinsic galaxy-to-galaxy variation and measurement error, with no residual trend in g_{bar} .

Figure 1 summarizes the comparison. Panel (a) shows D_{obs} against θ with the zero-parameter prediction $D = 1 + n_{\text{BE}}(\theta)$ overlaid; the threshold $(\ln 2, 2)$ is marked. Panel (b) shows the logarithmic residuals against $\log_{10} g_{\text{bar}}$ with $\pm\sigma = \pm 0.133$ dex bands. Panel (c) shows the occupation number $n = D - 1$ against θ , with binned medians superimposed on the individual points and the one-bit threshold $(\ln 2, 1)$ highlighted.

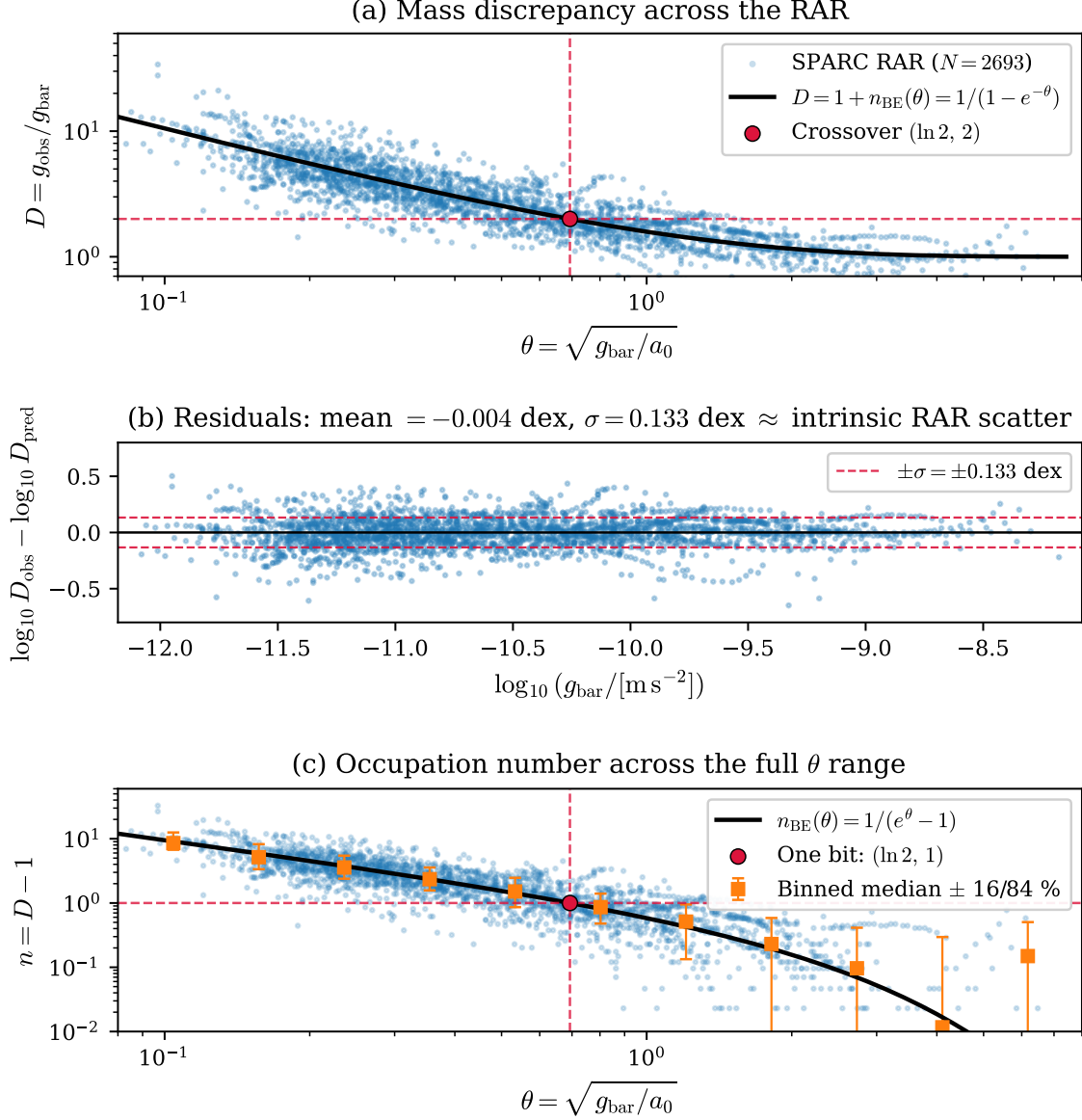


Figure 1: **The MOND interpolation function as a Bose–Einstein distribution over the SPARC RAR.** (a) Mass discrepancy $D = g_{\text{obs}}/g_{\text{bar}}$ against $\theta = \sqrt{g_{\text{bar}}/a_0}$ for all 2,693 data points in 153 SPARC galaxies [2]. The solid curve is the zero-parameter prediction $D = 1/(1 - e^{-\theta})$. The threshold at $\theta = \ln 2$, $D = 2$ is indicated. (b) Logarithmic residuals $\log_{10} D_{\text{obs}} - \log_{10} D_{\text{pred}}$ against $\log_{10} g_{\text{bar}}$. Mean residual = -0.004 dex, standard deviation 0.133 dex, in agreement with the McGaugh+2016 intrinsic RAR scatter of ~ 0.13 dex. Dashed bands mark $\pm\sigma$. (c) Entropy-mode occupation number $n = D - 1$ against θ , with binned medians (orange squares, $\pm 16/84$ th percentile) and the Bose–Einstein prediction $n_{\text{BE}}(\theta) = 1/(e^\theta - 1)$ (solid curve). The one-bit threshold $(\ln 2, 1)$ is marked. The data track the prediction more than two decades in n , from $n \sim 13$ to $n \sim 0.02$.

7.3 Occupation Number Across the Full θ Range

Table 1 tabulates the predicted and observed occupation numbers in ten theory-motivated diagnostic bins of θ spanning the full range calculated from the SPARC data, chosen to probe the qualitative regimes of the Bose–Einstein function including the analytic crossover at $\theta = \ln 2$.

Table 1: Predicted vs. observed Bose–Einstein occupation number across the SPARC RAR, in ten bins of $\theta = \sqrt{g_{\text{bar}}/a_0}$. N is the number of individual measurements in each bin. Predicted occupation: $n_{\text{BE}} = 1/(e^\theta - 1)$. Observed occupation: $n_{\text{obs}} = D_{\text{obs}} - 1$, median across the bin. The one-bit threshold ($\theta = \ln 2 = 0.693$) is the upper edge of the $\theta = 0.597$ bin, where $n_{\text{BE}} \simeq n_{\text{obs}} \simeq 1$.

θ	n_{BE} (predicted)	n_{obs} (median)	N
0.075	12.84	11.45	10
0.150	6.18	5.17	352
0.250	3.52	3.47	532
0.400	2.03	2.09	530
0.597	1.23	1.19	324
0.847	0.75	0.78	315
1.250	0.40	0.48	276
1.750	0.21	0.23	150
2.500	0.09	0.10	133
4.000	0.02	0.00	59

The Bose–Einstein prediction tracks the SPARC data from the deep-MOND regime at $\theta \sim 0.08$ ($n_{\text{BE}} \sim 13$) through the crossover threshold at $\theta \sim 0.69$ ($n_{\text{BE}} \sim 1$) into the Newtonian regime at $\theta \gtrsim 2$ ($n_{\text{BE}} \lesssim 0.1$), spanning occupation numbers from $n_{\text{BE}} \sim 13$ in the deep-MOND regime to $n_{\text{BE}} \sim 0.02$ in the Newtonian regime—a factor of more than 600—with no free parameters save for a_0 .

8 Discussion

8.1 Results

Starting from a complexified extension of the Einstein–Hilbert action with a scalar entropy potential in the imaginary sector, we have obtained: (i) the standard Einstein equations with $8\pi G$ coupling on the real sector, following from an equal-weight measure on the entropy axis; (ii) the decoupled massless scalar equation $\square\theta = 0$ for the entropy potential, with static spherically symmetric solution $\theta(r) = \sqrt{GM/a_0} r^{-1}$ identical to the empirical MOND parameter; (iii) the algebraic identity between the MOND interpolation function and the Bose–Einstein partition function over a single mode of dimensionless energy θ ; (iv) the location of the baryon-to-entropy threshold at $\theta = \ln 2$, corresponding to $g_{\text{bar}}/a_0 = (\ln 2)^2 = 0.4805$; and (v) an empirical validation against 2,693 SPARC data points with residual scatter of 0.133 dex, equal to the known intrinsic RAR scatter.

The construction has one free parameter beyond General Relativity: the acceleration scale a_0 , which enters through the identification (16). Both the interpolation function’s functional form and the threshold location are not fit, but derived.

8.2 The Binary Projection and the One-Bit Threshold

The numerical coincidence between the $\frac{1}{2}$ of Postulate 1 and the $\ln 2$ of the crossover threshold is not a coincidence. Both reflect the same binary structure of the projection. The equal-weight measure selects one of two sectors—real or imaginary—on each point of $\mathcal{M}_{\mathbb{C}}$; the resulting real-sector dynamics inherit a $\frac{1}{2}$ normalization. By Landauer’s principle, the minimum entropy cost of a binary selection is $k_B \ln 2$. The threshold at $\theta = \ln 2$ is the locus at which that cost first manifests as an excitation in the observable sector: exactly one bit’s worth of entropy-mode occupation, equal to one baryonic contribution. The projection and the threshold are algebraic duals.

8.3 Relation to Established Approaches

The demonstrated Bose–Einstein structure of the mass discrepancy places the result in direct conversation with several lineages.

Jacobson recovered the Einstein equations as an equation of state from the Clausius relation $\delta Q = T \delta S$ applied to local Rindler horizons [9]. Our construction complements this: where Jacobson requires thermodynamic input at the horizon, the present framework embeds a scalar entropy potential $\theta(x)$ globally on $\mathcal{M}_{\mathbb{C}}$ whose gradient is the local entropy current. The two approaches are complementary: Jacobson’s derivation requires thermodynamic input at a boundary, while the present framework provides the bulk statistical mechanics of the entropy sector that supplies that input.

Verlinde proposed gravity as an entropic force emergent from holographic entropy on screens [11], and subsequently derived MOND-like phenomenology from entropy displacement in de Sitter space, obtaining $a_0 \sim cH_0$ from the competition between area-law and volume-law entanglement [25], though without identifying the specific partition function. Our framework replaces the holographic screen with the imaginary sector of a complexified manifold and the emergent force with the Bose–Einstein occupation of entropy modes governed by $\square \theta = 0$. The scalar θ is not holographically bounded; it is a spacetime-filling field whose static spherical solution reproduces the MOND phenomenology without invoking a screen.

Recent work has sharpened the expectation that MOND admits a thermodynamic origin. Pikhitsa [26] derived the MOND equations from a thermodynamic treatment of gravitation in an expanding universe, obtaining $a_0 = 2cH$ without dark matter. Sheykhi and collaborators [27] have extended the correspondence between gravitational field equations and thermodynamics to construct relativistic MOND from modified entropic gravity with temperature corrections on the apparent horizon. Liu [28] studied Bose–Einstein condensates in MOND-inspired gravitational potentials, treating MOND as an input to condensate physics. The present result inverts this last direction: the MOND interpolation function is not a background for condensate dynamics but is itself the Bose–Einstein occupation statistics of the gravitational field.

Padmanabhan showed [10] that a wide class of gravitational actions admits a thermodynamic interpretation with an entropy functional on horizons. The present construction differs in that the entropy functional lives on the entropy axis $\mathcal{E}$ rather than being confined to horizons, and its excitations are directly observable as the galactic mass discrepancy.

More broadly, the present result demonstrates that entropic gravity is a genuinely nonlinear theory, showing how the entropy potential $\theta(r)$, sourced by the baryonic mass distribution through $\square\theta = 0$, couples back to observable dynamics through the Bose–Einstein partition function $Z(\theta) = 1/(1 - e^{-\theta})$, which is nonlinear in θ . In the deep-MOND regime ($\theta \ll 1$) the entropy sector dominates as $n_{\text{BE}} \sim 1/\theta$; in the Newtonian regime ($\theta \gg 1$) it freezes out as $n_{\text{BE}} \sim e^{-\theta}$. Matter sources entropy, entropy sources dynamics, and the coupling is exact and parameter-free.

We note that the present framework is consistent with recent cosmological tests of the gravitational force law. Gallardo et al. [8] used the kinematic Sunyaev–Zel’dovich effect to constrain the pairwise gravitational acceleration between galaxy clusters on scales of 30–230 Mpc, finding $g \propto 1/r^n$ with $n = 2.1 \pm 0.3$, consistent with Newtonian gravity and excluding the MOND force-law modification ($n = 1$) at 3.3σ . This result is expected in the present framework: the Einstein equations are recovered exactly on the real sector (Eq. 11), so the gravitational force law remains $1/r^2$ at all scales. The mass discrepancy attributed to dark matter arises not from a modification of the force law but from the thermal occupation of entropy modes governed by Bose–Einstein statistics. The Gallardo et al. result rules out MOND as a modified force law; it does not constrain MOND as a partition function over gravitational entropy modes, which preserves the standard force law by construction.

The standard view treats (1) as a useful fit function [5, 6] awaiting theoretical derivation. The near-zero intrinsic scatter and absence of residual correlations with any local or global galaxy property established by [2] sharpen this demand: the functional form is not approximately correct but exact to measurement precision across all morphological types, complete with no second parameter, no hidden variable, and no dependence on morphology, surface brightness, gas fraction, or radius beyond what the baryonic acceleration already encodes. The derivation presented here accounts for this completeness naturally: the entropy potential $\theta(r)$ is sourced entirely by the baryonic mass distribution through $\square\theta = 0$, so no additional galaxy property can enter. We have provided the derivation: the exponential is the Boltzmann factor of a single bosonic mode, the denominator is the single-mode partition function, and the argument of the exponential is the static spherical solution of a massless scalar equation on curved spacetime. The fit becomes a consequence of the underlying structure rather than an empirical input.

8.4 Open Questions

The acceleration scale a_0 is not derived within the present framework. This is a shared limitation with MOND itself, which treats a_0 as an empirical input [5, 6].

The identification (16) relies on the static, asymptotically flat, spherically symmetric solution $\theta = B/r$. The field equation $\square\theta = 0$ is general, but the functional form of θ for non-spherical or time-dependent sources requires further work. The static spherically symmetric case is the natural starting point precisely because the RAR data live there.

Postulate 1 remains the load-bearing physical input of the framework. It is formally

equivalent to a symmetry condition on the measure over the entropy axis and is analogous in status to Jacobson’s Clausius input and Verlinde’s holographic-entropy input. It is an open question whether it can be derived from a more primitive structural condition such as a path-integral measure, a consistency requirement on the complexified geometry, or a holographic principle. Postulate 1 also suppresses back-reaction at leading order; non-spherical and time-dependent extensions may require revisiting this assumption.

The entropy modes appearing in Eq. (22) have well-defined thermal occupation statistics and a clear interpretation as degrees of freedom traced out at $\tau = 0$ on the entropy axis $\mathcal{E}$. The present construction does not identify these modes with any specific microscopic object, and makes no claims about the short-distance or cosmological completion of the theory. Both identifications remain open questions whose resolution would require additional physical input beyond the scope of this paper.

9 Conclusion

The MOND interpolation function is a Bose–Einstein partition function. The “dark matter” contribution to galaxy dynamics is the mean thermal occupation of gravitational entropy modes, governed by the entropy potential $\theta(r) = \sqrt{GM/a_0} r^{-1}$ that arises as the static spherically symmetric solution of $\square\theta = 0$ on the real sector of a complexified Einstein–Hilbert action.

The transition from baryon-dominated to entropy-dominated dynamics occurs at $\theta = \ln 2$ —one bit of information, the Landauer cost [24] of the real-sector projection—corresponding to $g_{\text{bar}}/a_0 = (\ln 2)^2 = 0.4805$. Tested against 2,693 SPARC measurements in 153 galaxies, the prediction reproduces the mass discrepancy with residual scatter of 0.133 dex, equal to the intrinsic RAR scatter. No parameters are fit; a_0 is the only empirical input, and it is inherited from the existing MOND phenomenology.

Dark matter, in this picture (i.e., at the scale demonstrated over the course of this work), is not missing mass, but instead is the thermal population of the entropy sector of a complexified Einstein–Hilbert action, projected onto the real observational surface at $\tau = 0$, and bookkept as an excess acceleration by observers who have traced out the imaginary half of the geometry. The radial acceleration relation is the first consequence of this construction for which the data are sufficiently clean and complete to confirm that a parameter-free derivation from a Lagrangian is not merely possible but necessary: the baryons fully determine the dynamics, with no remainder, because the entropy potential sourced by the baryonic mass distribution has a unique solution. The generalization to time-dependent and non-spherical sources of the entropy field, and the derivation of a_0 within the framework, are the immediate open questions.

The mass discrepancy is the thermal occupation of the entropy sector of the complexified Einstein–Hilbert action, projected onto the real observational slice at $\tau = 0$:

$$D \equiv \frac{g_{\text{obs}}}{g_{\text{bar}}} = \underbrace{1}_{\text{baryons}} + \underbrace{n_{\text{BE}}(\theta)}_{\text{entropy}}, \quad \theta = \sqrt{\frac{g_{\text{bar}}}{a_0}} \quad (26)$$

Data and Code Availability

All analysis code, the RAR computation, and the reproduction script for Figure 1 are available at https://github.com/amr28693/einstein_16pi_MOND. The SPARC RAR dataset used for the empirical validation is publicly available from Lelli et al. [2].

Funding

This research received no external funding.

Conflicts of Interest

The author declares no conflicts of interest.

References

- [1] McGaugh, S.S.; Lelli, F.; Schombert, J.M. Radial Acceleration Relation in Rotationally Supported Galaxies. *Phys. Rev. Lett.* **2016**, *117*, 201101. <https://doi.org/10.1103/PhysRevLett.117.201101>.
- [2] Lelli, F.; McGaugh, S.S.; Schombert, J.M.; Pawlowski, M.S. One Law to Rule Them All: The Radial Acceleration Relation of Galaxies. *Astrophys. J.* **2017**, *836*, 152. <https://doi.org/10.3847/1538-4357/836/2/152>.
- [3] McGaugh, S.S. Milky Way Mass Models and MOND. *Astrophys. J.* **2008**, *683*, 137–148. <https://doi.org/10.1086/589148>.
- [4] McGaugh, S.S. The Third Law of Galactic Rotation. *Galaxies* **2014**, *2*, 601–622. <https://doi.org/10.3390/galaxies2040601>.
- [5] Milgrom, M. A Modification of the Newtonian Dynamics as a Possible Alternative to the Hidden Mass Hypothesis. *Astrophys. J.* **1983**, *270*, 365–370. <https://doi.org/10.1086/161130>.
- [6] Famaey, B.; McGaugh, S.S. Modified Newtonian Dynamics (MOND): Observational Phenomenology and Relativistic Extensions. *Living Rev. Relativ.* **2012**, *15*, 10. <https://doi.org/10.12942/lrr-2012-10>.
- [7] Milgrom, M. The MOND Limit from Spacetime Scale Invariance. *Astrophys. J.* **2009**, *698*, 1630–1638. <https://doi.org/10.1088/0004-637X/698/2/1630>.
- [8] Gallardo, P.A.; Pardo, K.; Philcox, O.H.E.; Battaglia, N.; Battistelli, E.S.; Bean, R.; Calabrese, E.; Choi, S.K.; Devlin, M.; et al. Test of the Gravitational Force Law on Cosmological Scales Using the Kinematic Sunyaev-Zel’dovich Effect. *Phys. Rev. Lett.* **2026**, *136*, 151002. <https://doi.org/10.1103/rk8v-rcm3>.

- [9] Jacobson, T. Thermodynamics of Spacetime: The Einstein Equation of State. *Phys. Rev. Lett.* **1995**, *75*, 1260–1263. <https://doi.org/10.1103/PhysRevLett.75.1260>.
- [10] Padmanabhan, T. Thermodynamical Aspects of Gravity: New Insights. *Rep. Prog. Phys.* **2010**, *73*, 046901. <https://doi.org/10.1088/0034-4885/73/4/046901>.
- [11] Verlinde, E. On the Origin of Gravity and the Laws of Newton. *J. High Energy Phys.* **2011**, *2011*, 29. [https://doi.org/10.1007/JHEP04\(2011\)029](https://doi.org/10.1007/JHEP04(2011)029).
- [12] Banerjee, R.; Majhi, B.R. Statistical Origin of Gravity. *Phys. Rev. D* **2010**, *81*, 124006. <https://doi.org/10.1103/PhysRevD.81.124006>.
- [13] Rubin, V.C.; Ford, W.K., Jr.; Thonnard, N. Extended Rotation Curves of High-Luminosity Spiral Galaxies. *Astrophys. J.* **1978**, *225*, L107–L111. <https://doi.org/10.1086/182804>.
- [14] Rubin, V.C.; Ford, W.K., Jr.; Thonnard, N. Rotational Properties of 21 SC Galaxies with a Large Range of Luminosities and Radii, from NGC 4605 ($R = 4$ kpc) to UGC 2885 ($R = 122$ kpc). *Astrophys. J.* **1980**, *238*, 471–487. <https://doi.org/10.1086/158003>.
- [15] Bertschinger, E., 2002. “Symmetry Transformations, the Einstein–Hilbert Action, and Gauge Invariance.” MIT OpenCourseWare Notes, 2002. Physics 8.962 Lecture Notes. <https://web.mit.edu/edbert/GR/gr5.pdf>
- [16] Misner, C.W.; Thorne, K.S.; Wheeler, J.A. *Gravitation*; W.H. Freeman: San Francisco, CA, USA, 1973.
- [17] Lang, S. *Algebra*, 3rd ed.; Graduate Texts in Mathematics; Springer: New York, NY, USA, 2002. <https://doi.org/10.1007/978-1-4613-0041-0>.
- [18] Zee, A. *Quantum Field Theory in a Nutshell*, 2nd ed.; Princeton University Press: Princeton, NJ, USA, 2010.
- [19] Bak, J.; Newman, D.J. *Complex Analysis*, 3rd ed.; Undergraduate Texts in Mathematics; Springer: New York, NY, USA, 2010. <https://doi.org/10.1007/978-1-4419-7288-0>.
- [20] Swendsen, R.H. *An Introduction to Statistical Mechanics and Thermodynamics*; Oxford University Press: Oxford, UK, 2012.
- [21] Griffiths, D.J. *Introduction to Electrodynamics*, 5th ed.; Cambridge University Press: Cambridge, UK, 2023. <https://doi.org/10.1017/9781009397735>.
- [22] Pathria, R.K.; Beale, P.D. *Statistical Mechanics*, 4th ed.; Academic Press: London, UK, 2021.
- [23] Unruh, W.G. Notes on Black-Hole Evaporation. *Phys. Rev. D* **1976**, *14*, 870–892. <https://doi.org/10.1103/PhysRevD.14.870>.

- [24] Landauer, R. Irreversibility and Heat Generation in the Computing Process. *IBM J. Res. Dev.* **1961**, 5, 183–191. <https://doi.org/10.1147/rd.53.0183>.
- [25] Verlinde, E.P. Emergent Gravity and the Dark Universe. *SciPost Phys.* **2017**, 2, 016. <https://doi.org/10.21468/SciPostPhys.2.3.016>.
- [26] Pikhitsa, P.V. MOND Reveals the Thermodynamics of Gravity. *arXiv:1010.0318v3* **2010**.
- [27] Sheykhi, A. MOND Theory and Thermodynamics of Spacetime. *arXiv:2510.14345* **2025**.
- [28] Liu, N. Ground State and Collective Modes of Bose–Einstein Condensates in Newtonian and MOND-inspired Gravitational Potentials. *arXiv:2601.01039* **2026**.

A Numerical Illustrations of the Projection Mechanism

Scope and interpretation of Appendix figures. The figures in this appendix are *schematic numerical illustrations* intended to visualize structural features of the real–imaginary decomposition and the real-sector projection defined in Section 2.3. They do *not* represent physical spacetime solutions of the field equations, do not constitute numerical integrations of the theory, and do not introduce an additional physical dimension. In particular, the entropy-axis coordinate τ is a statistical label rather than a spacetime coordinate, and quantities shown prior to projection are not directly observable. Observable curvature corresponds only to the projected real-sector component.

The projection onto the real sector constructed analytically in Section 2.3 makes three structural claims: that imaginary curvature is sourced by the entropy-field gradient, that the real and imaginary sectors carry equal kinetic weight prior to projection, and that the imaginary sector vanishes on the entropy-neutral slice $\tau = 0$ leaving observable curvature dominated by the real component. The three numerical demonstrations below illustrate each claim independently and are intended as conceptual visualizations of the projection mechanism, not as numerical evidence for the empirical results established in Section 7.

A.1 Imaginary Curvature as an Entropy Gradient

Consider a complexified metric perturbation $g_{\mu\nu} = \eta_{\mu\nu} + \epsilon e^{i\theta(x)}$, with $\theta(x)$ a local phase encoding entropy flow. To leading order in ϵ the imaginary part of the curvature scalar satisfies

$$\text{Im}(\mathcal{R}) \sim \nabla^\mu \theta \nabla_\mu \theta, \quad (27)$$

so that $\text{Im}(\mathcal{R})$ is sourced by the gradient structure of the entropy field. Figure 2 visualizes the real curvature (a Gaussian bump), the imaginary curvature from the phase field $\theta(x, y) = \arctan2(y, x)$, and the combined complexified curvature magnitude $|\mathcal{R}^{\mathbb{C}}|$ prior to projection. The real sector inherits the symmetry structure through projection of Section 2.3.

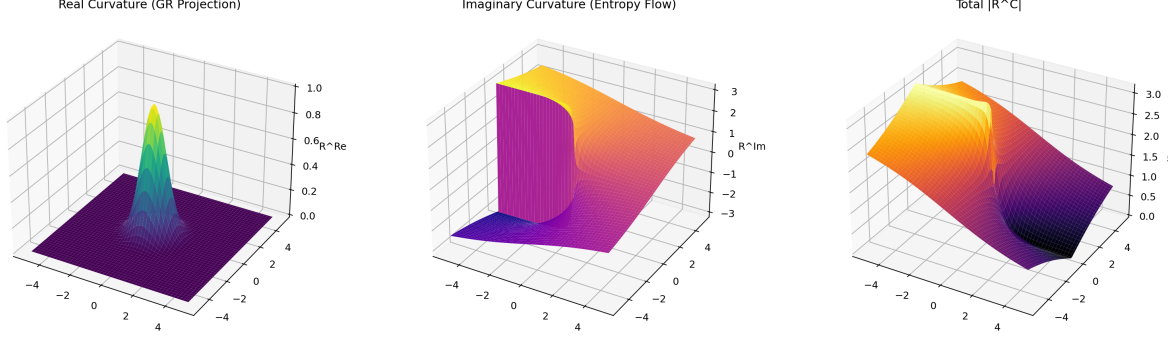


Figure 2: Illustrative decomposition of complexified curvature over a 2D spatial slice. **(Left)** Real curvature component (Gaussian profile). **(Centre)** Entropy phase field $\theta(x, y) = \arctan2(y, x)$, whose gradient structure sources the imaginary curvature sector. **(Right)** Magnitude of the algebraic superposition $|\mathcal{R}^{\mathbb{C}}|$ *prior to projection*. The combined quantity is shown solely to visualize the symmetry structure between sectors assumed in Postulate 1; observable curvature corresponds only to the projected real-sector component.

A.2 Equal-Weight Symmetry Between Real and Imaginary Sectors

To examine the gradient structure of the complexified field directly, dual real and imaginary Gaussian soliton profiles are defined on a 2D grid ($[-5, 5]^2$, 200×200), with spatial gradients computed via central finite differences and smoothed with a Gaussian filter ($\sigma = 1.0$). Table 2 lists gradient-magnitude statistics for the real sector alone, the imaginary sector alone, and the combined field.

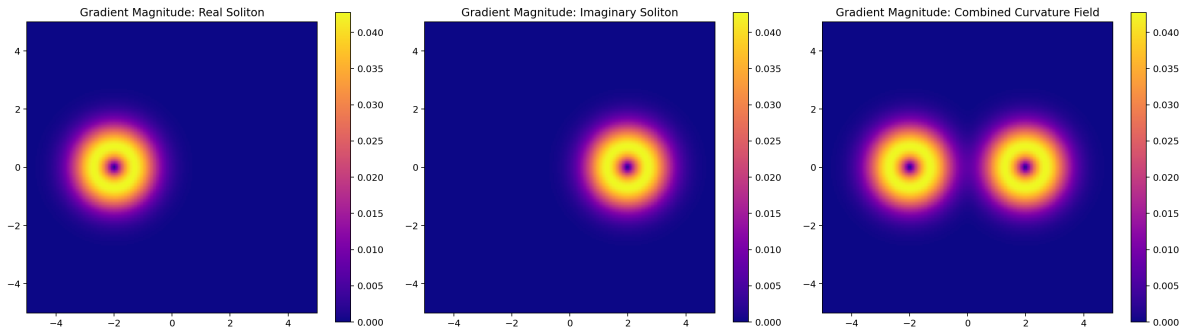


Figure 3: Gradient magnitudes for illustrative real, imaginary, and combined soliton profiles. **(Left)** Real-sector soliton. **(Centre)** Imaginary-sector soliton. **(Right)** Algebraic superposition of real and imaginary components prior to projection. This figure is included solely to demonstrate the equal-weight symmetry between sectors assumed in Postulate 1; the combined field visualized here is not an observable quantity and does not represent physical spacetime curvature.

Table 2: Gradient magnitude statistics for the real, imaginary, and combined soliton profiles of Figure 3. The real and imaginary components have identical statistics—the numerical signature of the equal-weight symmetry assumed in Postulate 1. The visualized combined field shows larger mean and standard deviation from constructive superposition.

Field	Min	Max	Mean	Std. Dev.
Real gradient	0.00000	0.04270	0.00276	0.00836
Imaginary gradient	0.00000	0.04270	0.00276	0.00836
Combined gradient	0.00000	0.04271	0.00552	0.01118

A.3 Entropy-Driven Projection at $\tau = 0$

Define a complex scalar field $\Psi(x, \tau) = A(x, \tau) e^{i\theta(x, \tau)}$ with Gaussian envelope $A = e^{-(x^2 + \tau^2)/2}$ and entropy-modulated phase $\theta = \sin(2\pi x) \tanh(\tau)$, on a 300×300 grid over $[-5, 5]^2$. The entropy density $|\nabla\theta|^2$ is smoothed by a Gaussian filter. At $\tau = 0$, the field is projected onto the spatial axis and curvature proxies are computed as

$$\mathcal{C}_{\text{real}}(x) = \left| \frac{d}{dx} \text{Re}[\Psi(x, 0)] \right|^2, \quad \mathcal{C}_{\text{imag}}(x) = \left| \frac{d}{dx} \text{Im}[\Psi(x, 0)] \right|^2, \quad \mathcal{C}_{\text{total}} = \mathcal{C}_{\text{real}} + \mathcal{C}_{\text{imag}}.$$

Figure 4 displays a visualization of the real and imaginary parts of Ψ , the entropy density, and the projected curvature at $\tau = 0$. The projected curvature is dominated by the real component; the imaginary contribution vanishes on the entropy-neutral slice, consistent with the framework’s structural prediction that the real-sector projection captures the observable curvature.

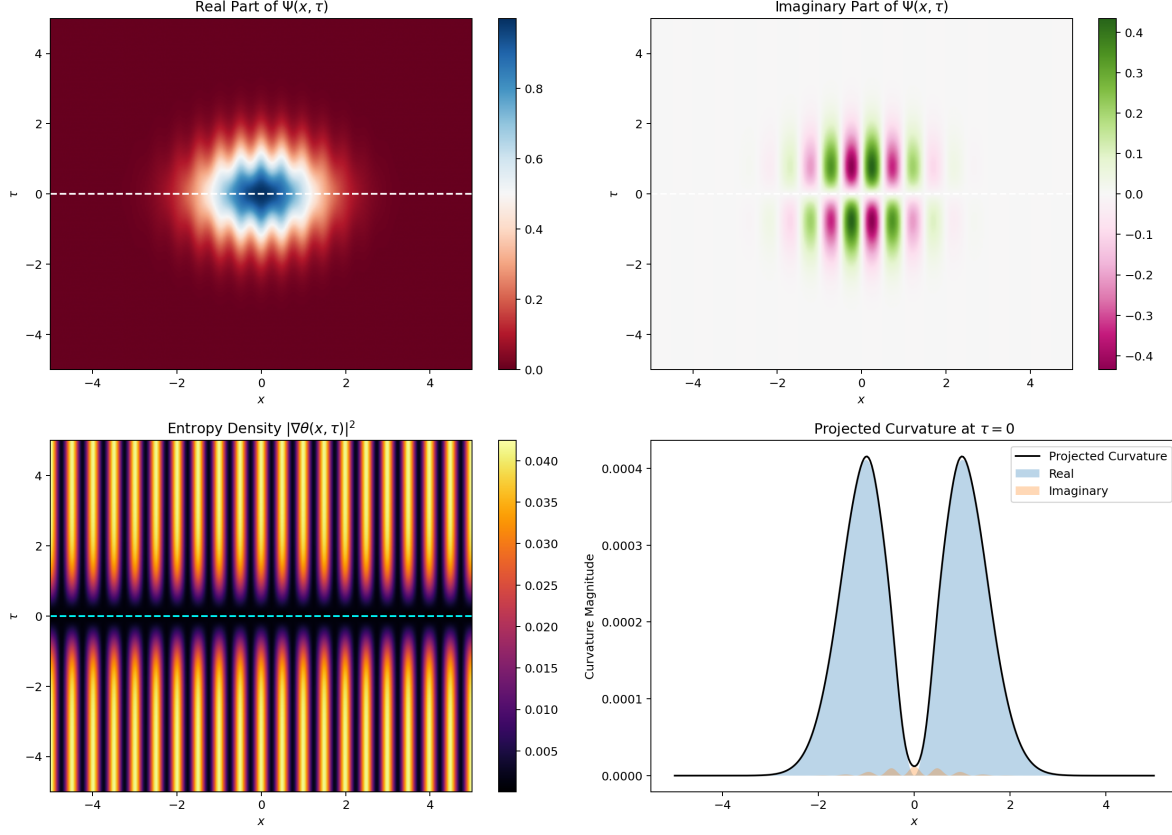


Figure 4: Schematic numerical illustration of the real-sector projection. (**Top left**) $\text{Re}[\Psi(x, \tau)]$. (**Top right**) $\text{Im}[\Psi(x, \tau)]$. (**Bottom left**) Entropy density $|\nabla\theta|^2$. (**Bottom right**) Curvature proxy after application of the real-sector projection, which for reflection-symmetric configurations is equivalent in this illustration to evaluation at the entropy-neutral point $\tau = 0$. The τ coordinate is a statistical label rather than a physical dimension; no evolution in τ is implied, and only the projected real component corresponds to observable curvature.